Circuits

ALL ELECTRONIC AND ELECTRICAL SYSTEMS AND EQUIPMENT ARE BUILT FROM CIRCUITS.

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Circuits are composed of power sources, conductors, and electronic or electrical components that carry out specific tasks.

Circuit basics

In any circuit, a power source—such as a cell—pushes electrical current along one or more conductors, often wires. When the current passes through a component, such as a light bulb, the component changes the electricity and also changes itself in response. For example, a resistor controls the flow of current to protect the device from overload. Similarly, a light bulb opposes the current and lights up. If the circuit is broken—by means of a switch, for example—the current ceases to flow.

△ **Switch**
This allows or halts the flow of current.

△ **Cell**
This causes current to flow around the circuit.

△ **Capacitor**
This device stores electrical charge.

△ **Light bulb**
This will light up when current flows through it.

△ **Voltmeter**
This device measures the voltage, in volts.

△ **Ammeter**
This component measures the current, in amps.

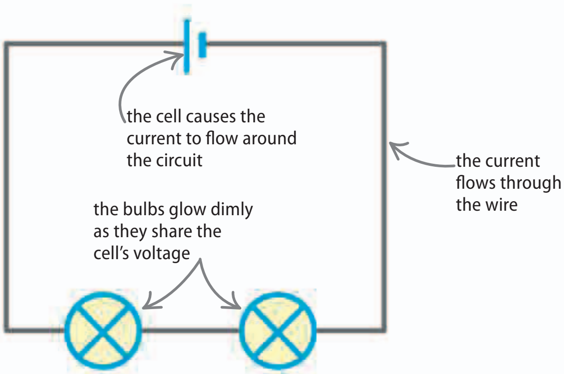
△ **Resistor**
The purpose of this is to resist the flow of current.

△ **Variable resistor**
This device controls the amount of current.

△ **Motor**
A motor moves when current flows through it.

In series

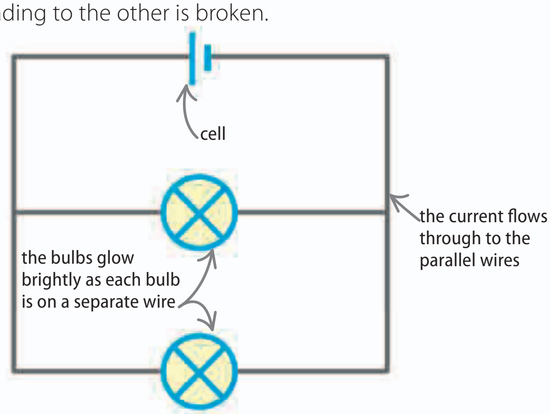
When components are connected in series, they share the voltage of the power source, such as a cell. If there are two identical components, then each will receive half the voltage. The current will stop flowing around the circuit if there is a break at any point.



△ **Series circuit**
These two bulbs are arranged in series so they have to share the voltage. They glow dimly as a result.

In parallel

When components are connected in parallel, they are each subject to the whole voltage from the power source. The current will continue to flow through one bulb if the wire leading to the other is broken.



△ **Parallel circuit**
These two bulbs are arranged in parallel so they both receive the full voltage from the cell and glow brightly.